Transit Equity and Built Environment Disparities: Omaha Metro vs. Council Bluffs

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1 Smart Location Database

1.1 1.1 Variables

- **Auto Regional Job Accessibility (`D5AR`):** Total regional employment accessible within a 45-minute travel time by automobile from the census block group. Higher values indicate greater regional labor-market reach via personal vehicle.
- **Census Block Group Identifier (`GEOID` / `GEOID12`):** Unique 12-digit Federal Information Processing Series (FIPS) code used to uniquely identify each Census Block Group nationwide and link tabular data to Census TIGRIS shapefiles.
- **Distance to Nearest Transit Stop (`D4A`):** Network pedestrian walking distance (in meters) from the block group population centroid to the nearest fixed-route public transit stop (capped at an upper threshold of 3,000 meters to limit the distortion of unserved areas).
- **Employment & Housing Mix (`D2B_E8MIX`):** Balance between housing units and employment opportunities within the block group. Scale: 0 = pure residential or commercial; 1 = highly balanced land-use mix.
- **Gross Employment Density (`D1C`):** Total employment (jobs) per acre of unprotected land. Higher values indicate greater concentrations of commercial activity.
- **Gross Residential Density (`D1A`):** Housing units per acre of unprotected land. Higher values reflect denser residential settlement patterns capable of supporting frequent transit.
- **Jobs-to-Household Ratio (`D2A_JPHH`):** The ratio of local jobs to resident households within a block group, capturing localized mixed-use balance and proximity between trip origins and destinations.
- **Median Household Income (`MedIncome`):** Median household income within the census block group reported in USD.

- **National Walkability Index (`NatWalkInd`):** Composite index evaluating overall pedestrian friendliness based on density, diversity, design, and transit proximity. Scale: 1 = Least Walkable (Auto-Dependent Car Suburb); 20 = Most Walkable (Pedestrian-Oriented Urban Core).
- **Percentage of Low-Income Households (`Pct_LowInc`):** Proportion of block group households living below the regional low-income threshold.
- **Percentage of Zero-Car Households (`Pct_ZeroCar`):** Proportion of occupied housing units within the block group that do not own or have access to a personal vehicle, serving as the operational indicator of transit dependency.
- **Street Intersection Density (`D3A / D3B`):** Density of multi-leg pedestrian street intersections per square mile of unprotected land. Higher values reflect a tighter, more interconnected street grid that minimizes pedestrian detour distances.
- **Transit Regional Job Accessibility (`D5CR`):** Total regional employment accessible within a 45-minute commute via public transit, capturing functional network utility beyond physical stop presence.

2 2 Methods

2.1 2.1 Council Bluffs - Omaha Comparisons

This analysis compared block groups across the Omaha-Council Bluffs metropolitan region. Eight measures were extracted from the U.S.Environmental Protection Agency's Smart Location Database (SLD Version 3.0) to capture key dimensions of urban form, pedestrian infrastructure, and transit accessibility. EPA measures were standardized with a 12-digit Census Bureau FIPS identifier and linked to 2010 cartographic shapefiles from the U.S. Census bureau using the tigris spatial library in R.

Environmental indicators in metropolitan areas typically exhibit skew and non-normality due to jobs and housing being distributed unevenly within cities. Distribution traits were checked to determine degree of skew and kurtosis in order to identify if parametric tests would be appropriate to use on the data.

As a result of this trend Omaha and Council bluffs were evaluated using a two-tailed Mann-Whitney U test based on median rather than sample means. To reduce Type I error inflation from repeated hypothesis testing a Bonferroni adjustment was applied. In order to determine practical significance, an effect size of Cliff's delta (δ) was obtained alongside 95% confidence intervals.

2.2 2.2 Bivariate Transit Equity and Spatial map

Based on findings of the hypothesis testing results, spatial mapping was conducted to identify disparities across state lines. The goal was to map transit need against existing transit supply with respect to the border between Nebraska and Iowa. Two separate variables were used as evidence of transit supply. Physical stop proximity (D4A), which looked at the physical distance to the nearest transit stop and system destination reach (D5CR) which measured the volume of jobs within a 45 minute transit trip. These produced a 9-cell matrix ranging from high access/high car ownership to critical areas with low transit access and low car ownership. Because approximately half of regional block groups lack fixed-route transit coverage and share the 3,000-meter cap, stop proximity tiers were assigned by grouping all unserved block groups into the lowest access tier, with the remaining served block groups partitioned by median walking distance. These terciles were split with the lowest third being low car ownership. This was to determine if lack of transit stops or other factors explained the difference in transit access to jobs.

3 3 Results:

3.1 3.1 Metropolitan Disparities and Differences Between Metros

Are there differences in transit access between Omaha and Iowa metro areas and are they explained by typical urban population gradients or as a result of the boundary between the two states?

Figure 1: Comparative Density Distributions of Transit Equity Indicators

Kernel density estimates across census block groups in Omaha (NE) and Council Bluffs (IA)

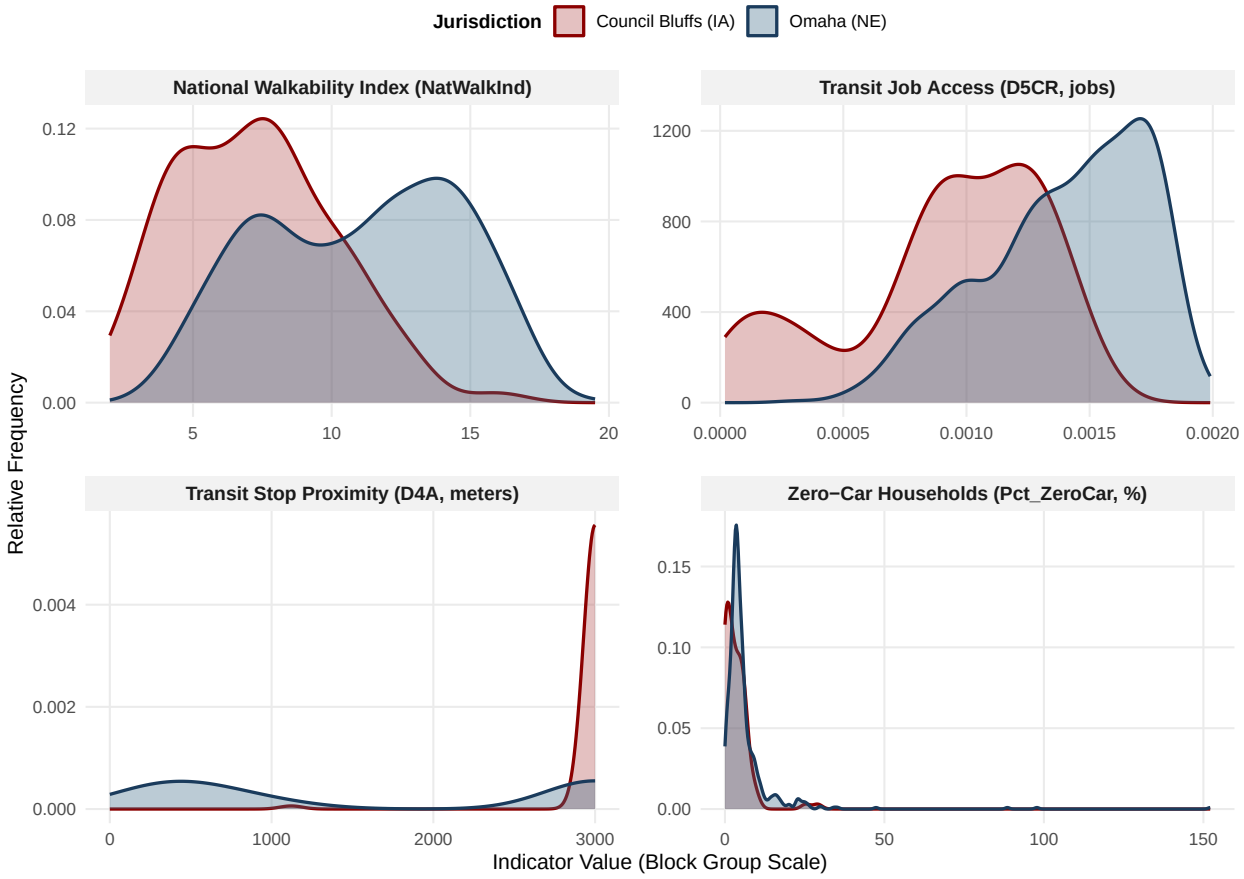


Figure 1: Distribution profiles of 8 indicators between both Council Bluffs and Omaha to display skew and normality. Figure 1: Comparative Density Distributions of Built Environment and Transit Equity Indicators Between Council Bluffs (IA) and Omaha (NE).

Across the bi-state metropolitan area, transit access and urban infrastructure have uneven patterns. Rather than a gradient that peaks at the metropolitan core and then has a smooth decline the further you get away from downtown there is an uneven distribution of connectivity and transit services that drops off starkly at the Missouri river boundary. Findings from these distributions (figure 1) show Council Bluffs has lower intersection density and more dispersed street pattern that would be consistent with car based development. the distribution for car ownership shows the majority of households have at least one vehicle. However a small collection of block groups have disproportionately high concentrations of zero-vehicle households. Additionally job access by car is tightly grouped around the interstate network.

Analysis across the bi-state area confirms inequity exists between Omaha and Council Bluffs across seven of the eight indicators. The only exception was there was an even number of jobs to households on both sides. Omaha block groups had significantly greater transit employment reach, superior walkability, broader automobile job access, and far closer proximity to physical transit stops. Conversely Council Bluffs has a lower proportion of zero-car households indicating that residents are reliant on car ownership for mobility. Additionally Council Bluffs resident despite having a higher proportion of zero car households have fewer access to jobs than their Omaha peers. For those without a vehicle in council bluffs they are even more vulnerable with longer walking distances and restricted access to jobs via transit.

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## Successfully matched block groups: 727
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Table 1: Table 2: Non-Parametric Mann-Whitney U Tests and Cliff's Delta Effect Sizes (Bonferroni-Adjusted)

Indicator	Dimension	Omaha Median	Council Bluffs Median	W Statistic	Bonferroni p-value	Sig.
NatWalkInd	Composite Walkability Index	11.33	7.17	12661	< 0.0001	****
D1A	D1: Residential Density (Units/Acre)	2.75	1.66	20586	< 0.0001	****
D2A_JPHH	D2: Diversity (Jobs/HH Ratio)	0.33	0.39	30595	1	ns
D3A	D3: Design (Intersection Density)	21.13	14.59	21982	0.0006	***
D4A	D4: Transit Proximity (Stop Distance, m)	873.74	3000.00	46122	< 0.0001	****
D5AR	D5: Auto Dest. Accessibility (45m Jobs)	81656.00	52881.00	10294	< 0.0001	****
D5CR	D5: Transit Dest. Accessibility (45m Jobs)	0.00	0.00	10294	< 0.0001	****
Pct_ZeroCar	Auto: Zero-Vehicle Households (%)	4.12	2.76	20045	< 0.0001	****

3.2 3.2 Transit Need vs Job Accessibility

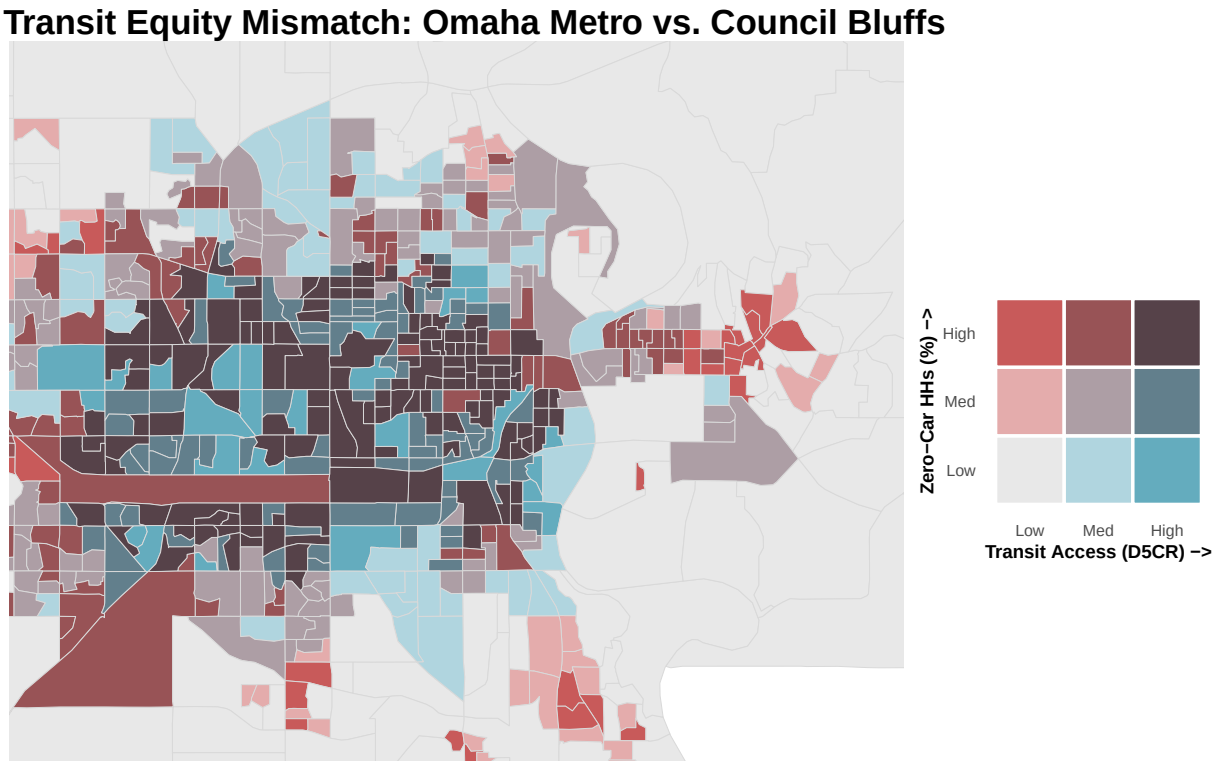


Figure 2: Spatial mismatch between zero-vehicle household and job accessibility.

Figure 2 visualizes transit equity between the two states by cross referencing percentage of households with no car against the functional transit supply (the number of regional jobs accessible via transit). Critical transit deserts are the bright red blocks with high proportion of zero car households as well as low amounts of transit accessible jobs. In Council Bluffs, these areas are neighborhoods bordering the river. In Omaha there are some of these in portions of north and south Omaha. Residents living in this area face significant structural challenges by not owning a vehicle. In contrast the dark red can be seen along Omahas Dodge street and represents those that have low car ownership but have a high amount of jobs accessible via transit. This means those without cars in these areas have alternative ways off accessing job opportunity. There is very little blocks of this color in Council Bluffs who lack transit options to buffer the effects of low car ownership.

3.3 3.3 Physical Transit Stop Proximity Role in Job Access

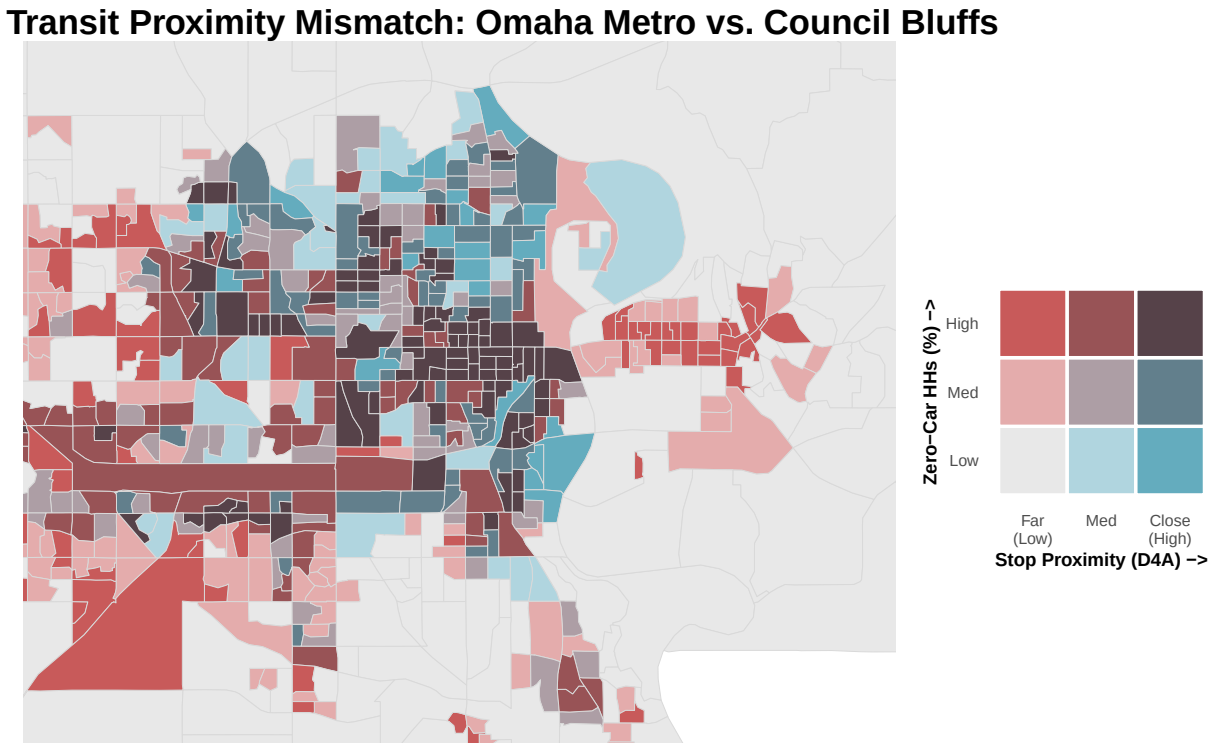


Figure 3: Walking proximity and percent of zero car households. Figure 3 evaluates physical infrastructure coverage by mapping transit need (percentage of zero-car households) against walking distance to the nearest transit stop. The Bright red represents those that have walking distance to transit stops that exceed 3000 meters and have a higher percentage of zero-vehicle households. In these groups access to transit stops is poor making them part of a transit desert. Nearly 40% of Council Bluffs transit sits beyond walking distance for those living in Council Bluffs. The absence of transit stops creates a barrier for residents and a greater need for vehicle ownership. This suggests that a great deal of the problem for Council Bluffs is a result of physical infrastructure.

4 Policy and Planning recommendations

Council Bluff has an access cliff at the state line where bus services are limited and reliance on cars is much greater when compared to Omaha. Council Bluffs resident rely on jobs located within the core metro area of Omaha but are cut off due to poor transit services and lack of a vehicle. An interstate partnership from Council Bluffs to key job areas in Omaha is needed to bridge the job accessibility gap that exists between the two cities. Additionally land use reforms in commercial areas within Council Bluffs to promote greater walkability through more protected pedestrian crossings and continuous sidewalk networks.This would also reduce walking barriers to transit hubs in areas with higher population density. Additionally a shared bus pass system between Omaha and Council Bluffs may help to reduce friction between the two systems as well as make it more realistic for individuals to transfer from different bus routes without long wait times.

5 4 Limitations

The distance-to-transit-stop variable (D4A) was assigned an artificial upper threshold of 3,000 meters in the source dataset. While this cap limits distortion from extreme rural outliers, it potentially masks the true degree of physical isolation experienced by households situated beyond that perimeter. Similarly, the 45-minute commute window (D5CR, D5AR) applies a fixed ceiling that may not reflect the reality of Council Bluffs residents who must undertake daily interstate commutes well beyond 45 minutes to reach regional employment centers.